Abalone Age Prediction - Complete Model Reference

Summary Statistics

Category	Count	Description
Standard Linear Models	11	Baseline through kitchen sink - progressively adding features to standard OLS regression
LM + Interactions	4	Log-transformed models with explicit interaction terms between key predictors
Regularized LM	3	Penalized regression (Ridge, Lasso, Elastic Net) to prevent overfitting
Nonlinearity Fix	9	Models addressing non-linear relationships via splines, transforms, and polynomials
Enhanced Interactions	8	Extended interaction models testing additional predictor combinations
Enhanced Regularized	3	Regularized models with expanded feature set including all interactions
Ensemble Models	5	Combinations of multiple models via averaging or weighted averaging
Total Models	**43**	38 single models + 5 ensembles

---

1. STANDARD LINEAR MODELS (M1-M11)

Model	Name	Response	Key Features	Description
M1	Baseline	Age	Sex, Length, Diameter, Height, Weight, Shucked Weight, Viscera Weight, Shell Weight	Original 8 features only - baseline for comparison
M2	Simple	Age	is_infant, size_mean, Weight, shell_thickness_proxy	Basic engineered features - simplified model
M3	Ratios	Age	M2 + shell_ratio, shucked_ratio	Added weight ratio features
M4	BioFlag	Age	M3 + is_slow_growth	Added biological anomaly flag
M5	Polynomial	Age	M4 + I(size_mean^2)	Added quadratic term for non-linearity

Model	Name	Response	Key Features	Description
M6	Interaction	Age	M5 + size_mean:Weight	Added size-weight interaction term
M7	Log	log(Age)	is_infant, size_mean, Weight, shell_thickness_proxy, shell_ratio, shucked_ratio, is_slow_growth	Log transform on response variable
M8	Log_Poly	log(Age)	M7 + I(size_mean^2)	Log response + quadratic size term
M9	Log_Log	log(Age)	is_infant, log(size_mean), Weight, shell_thickness_proxy, shell_ratio, is_slow_growth	Log-log transformation
M10	TopCorr	Age	is_infant, shell_thickness_proxy, Height, size_mean, Shell Weight, density	Top correlated features only
M11	KitchenSink	Age	Many features including viscera_ratio, density, volume_ellipsoid, height_diameter_ratio	Many features combined - tests multicollinearity

---

2. LM + INTERACTIONS (M12-M15)

Model	Name	Response	Key Features	Description
M12	Log_MoreInt	log(Age)	M8 + size_shell_interaction, weight_infant_interaction	Log model with 2 key interactions
M13	Log_Cubic	log(Age)	M8 + I(size_mean^3)	Log model with cubic polynomial term
M14	Log_Density	log(Age)	M8 + density_size_interaction	Log model with density-size interaction
M15	Log_FullInt	log(Age)	M8 + size_shell_interaction, size_shucked_interaction, weight_infant_interaction, shell_infant_interaction	Log model with 4 key interactions - **top performer**

---

3. REGULARIZED LINEAR MODELS (M16-M18)

Model	Name	Response	Regularization	Description
M16	Ridge	log(Age)	L2 (alpha=0)	Shrinks coefficients toward zero, keeps all predictors
M17	Lasso	log(Age)	L1 (alpha=1)	Performs feature selection by zeroing coefficients
M18	ElasticNet	log(Age)	L1+L2 (alpha=0.5)	Hybrid combining Ridge and Lasso benefits

Formula used: `log(Age) ~ is_infant + size_mean + size_mean_squared + Weight + shell_thickness_proxy + shell_ratio + shucked_ratio + is_slow_growth + size_shell_interaction + size_shucked_interaction + weight_infant_interaction + density + volume_ellipsoid + height_diameter_ratio`

---

4. NONLINEARITY FIX MODELS (M19-M27)

Model	Name	Response	Key Features	Description
M19	LogWeights	log(Age)	is_infant, size_mean, log_weight, log_shell_weight, shell_ratio, shucked_ratio, is_slow_growth	Log transforms on weight predictors
M20	AllLog	log(Age)	is_infant, log_size_mean, log_weight, log_shell_weight, log_shucked_weight, is_slow_growth	All predictors log-transformed
M21	Spline3	log(Age)	is_infant, ns(size_mean, df=3), Weight, shell_thickness_proxy, shell_ratio, shucked_ratio, is_slow_growth	Natural spline with 3 degrees of freedom
M22	Spline4	log(Age)	ns(size_mean, df=4)	Natural spline with 4 degrees of freedom
M23	Spline_Both	log(Age)	is_infant, ns(size_mean, df=3), ns(Weight, df=3), shell_ratio, shucked_ratio, is_slow_growth	Splines on both size and weight
M24	BoxCox	Age_bc	is_infant, size_mean, I(size_mean^2), Weight, shell_thickness_proxy, shell_ratio, shucked_ratio, is_slow_growth	Box-Cox transformed response
M25	Spline_LogWt	log(Age)	is_infant, ns(size_mean, df=3), log_weight, log_shell_weight, shell_ratio, shucked_ratio, is_slow_growth, weight_infant_interaction	Spline + log weights + interaction

Model	Name	Response	Key Features	Description
M26	Quartic	log(Age)	I(size_mean^4) included	Quartic (4th degree) polynomial
M27	Sqrt	log(Age)	sqrt_size_mean, sqrt_weight, sqrt_shell_weight	Square root transforms as alternative

5. ENHANCED INTERACTION MODELS (M28-M35)

Model	Name	Response	Added Features	Description
M28	DensityInt	log(Age)	M15 + density_infant_interaction	Tests density effect specific to infants
M29	WeightRatioInt	log(Age)	M15 + weight_shell_interaction, weight_shucked_interaction	Tests weight-ratio combinations
M30	SlowGrowthInt	log(Age)	M15 + slow_growth_size	Tests slow growth interaction with size
M31	InfantSizeSq	log(Age)	M15 + infant_size_squared	Tests quadratic size effect for infants
M32	CubicInt	log(Age)	M15 + I(size_mean^3)	Cubic polynomial with interactions
M33	KitchenSinkInt	log(Age)	M15 + density_infant_interaction, slow_growth_size, infant_size_squared	Many interactions combined
M34	SplineInt	log(Age)	ns(size_mean, df=3) + 4 interactions	Spline flexibility with interactions
M35	LogWeightInt	log(Age)	log_weight instead of Weight + 4 interactions	Log weight with interactions

6. ENHANCED REGULARIZED MODELS (M36-M38)

Model	Name	Response	Regularization	Description
M36	Ridge_Enh	log(Age)	L2 (alpha=0)	Ridge with all enhanced features and interactions
M37	Lasso_Enh	log(Age)	L1 (alpha=1)	Lasso with all enhanced features and interactions
M38	ElasticNet_Enh	log(Age)	L1+L2 (alpha=0.5)	Elastic Net with all enhanced features

Enhanced formula includes: size_mean_cubed, viscera_ratio, size_viscera_interaction, density_infant_interaction, slow_growth_size, infant_size_squared, weight_shell_interaction

---

7. ENSEMBLE MODELS (E1-E5)

Model	Name	Components	Method	Description
E1	Top3_Orig	M7, M8, M9	Simple average	Average of top 3 original log models
E2	Nonlinear	M21, M22, M25	Simple average	Average of best nonlinearity fix models
E3	Mix	M15, M37, M38	Simple average	Mix of best from different categories
E4	All	All log models	Simple average	Average of all log-transformed models
E5	Weighted	M15, M37, M38	Weighted by 1/MAE	Inverse MAE weighting for best performers

---

MODEL PERFORMANCE SUMMARY

Rank	Model	Type	MAE	Description
1	E5_Weighted	Ensemble	~1.34	Weighted ensemble of top performers
2	M15_Log_FullInt	LM + Interactions	~1.35	Log model with 4 key interactions
3	M37_Lasso_Enh	Enhanced Regularized	~1.36	Enhanced Lasso with feature selection
4	M38_ElasticNet_Enh	Enhanced Regularized	~1.36	Enhanced Elastic Net

Rank	Model	Type	MAE	Description
5	M8_Log_Poly	Standard LM	~1.37	Log response + quadratic term

---

KEY INSIGHTS

1. **Log transformation of Age is essential** - All top models use log(Age) as response
2. **Interactions provide biggest gains** - M15 with 4 interactions outperforms simpler models
3. **Regularization helps** - Prevents overfitting with many features
4. **Ensembles provide modest improvement** - ~0.01 MAE gain over best single model
5. **Splines offer flexibility** - But don't significantly outperform polynomial terms
6. **is_infant is critical** - Appears in all top models

R CODE: Best Model (M15)

r

```r
m15 <- train(
  log(Age) ~ is_infant + size_mean + I(size_mean^2) + Weight +
    shell_thickness_proxy + shell_ratio + shucked_ratio + is_slow_growth +
    size_shell_interaction + size_shucked_interaction +
    weight_infant_interaction + shell_infant_interaction,
  data = abalone,
  method = "lm",
  trControl = trainControl(method = "cv", number = 10)
)
```